Cylinder with heater

W21 (2021) — English translation

A horizontally located cylindrical vessel of volume $2V_0$ is divided into two equal parts by a movable piston and filled with a monatomic ideal gas. The piston moves without friction. The left wall, the side surface of the entire vessel and the piston do not conduct heat. The temperature of the right wall is maintained constant and equal to the initial gas temperature T_0 in both parts of the vessel. The initial gas pressure in both parts is p_0. After turning on the heater on the left side of the vessel, the piston begins to move slowly.

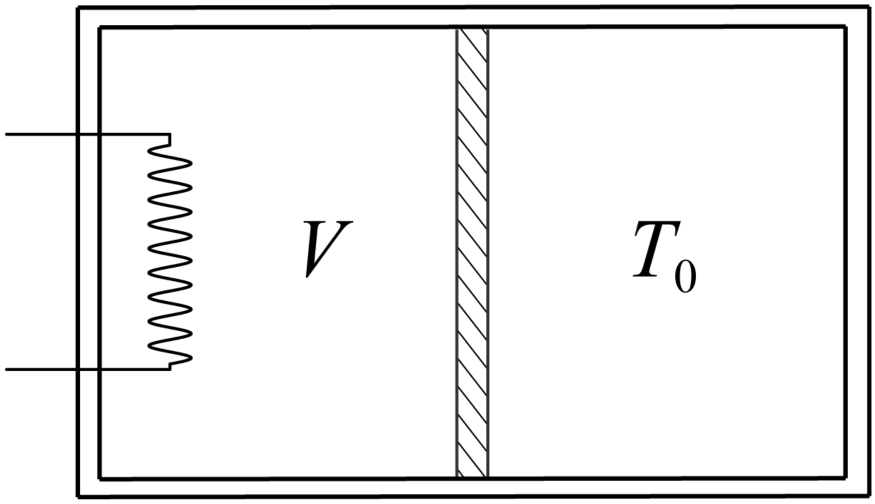

Figure 1: Figure 1.

A1	Determine the dependence of gas pressure p on temperature T in the left part of the vessel.	**3.00**

A2	For this process, determine how the heat capacity of the gas on the left side of the vessel depends on its volume V.	**7.00**

Source: `https://pho.rs/p/67`